4 **(a)** **(i)** effective resistance across BN $= \left(\frac{1}{2R} + \frac{1}{R+R}\right)^{-1} = R$

effective resistance across AM $= \left(\frac{1}{2R} + \frac{1}{R+R_{BN}}\right)^{-1} = \left(\frac{1}{2R} + \frac{1}{R+R}\right)^{-1}$ M1

$= R$ A0

or effective resistance across AM $= \left(\frac{1}{2R} + \frac{1}{R+\left(\frac{1}{2R}+\frac{1}{R+R}\right)^{-1}}\right)^{-1}$ (M1)

(ii) use potential divider principle, $V_{AM} = \frac{R}{R+R} \times V = \frac{1}{2}V$ M1

$V_A - V_M = V_A - 0 = V_{AM}$, so $V_A = \frac{1}{2}V$ A1

(iii) $V_D = \frac{1}{2}V_C = \frac{1}{2} \times \frac{1}{2}V_B = \frac{1}{4} \times \frac{1}{2}V_A = \frac{1}{8} \times \frac{1}{2}V$ C1

$V_D = \frac{1}{16}V$ A1

(b) *either* $R_P = \frac{6}{6\times10^{-3}}$ $(= 1000\ \Omega)$ C1

$R_Q = \frac{6}{3\times10^{-3}}$ $(= 2000\ \Omega)$ C1

$R_{MN} = \left(\frac{1}{1000} + \frac{1}{2000}\right)^{-1} = 667\ \Omega$ A1

or p.d. across MN = 6 V (C1)

total current = 6 + 3 = 9 mA (C1)

$R = \frac{6}{9\times10^{-3}} = 667\ \Omega$ (A1)

5 (a) (i) region of space which a force is felt/experienced B1